

Workforce AI + Communication + Studio Spec Pack

Reference guide for the packaged architecture specs, usage order, and repo-install workflow.

What this pack is for

This pack defines how Workforce can add OpenAI-backed AI capabilities without collapsing module boundaries. It separates conversation transport, AI orchestration, and structured Studio design logic so the platform can grow cleanly.

Included files

README.md

Top-level overview of the pack and recommended reading order.

specs/AI_GATEWAY_MODULE_SPEC.md

Shared AI orchestration module spec for provider configuration, model routing, prompts, tools, parsing, and request logs.

specs/COMMUNICATION_MODULE_SPEC.md

Reusable interaction infrastructure spec for threads, messages, participants, attachments, and message links.

specs/STUDIO_AI_INTEGRATION_SPEC.md

Studio integration spec describing how Studio uses Communication plus AI Gateway without losing ownership of structured design state.

copilot_install/COPILOT_INSTALL_AI_COMM_STUDIO_SPEC_PACK.md

Human-readable guide for placing the files into an existing Workforce repo.

copilot_install/COPILOT_INSTALL_AI_COMM_STUDIO_SPEC_PACK_PROMPT.md

Copilot-ready install prompt for adding the files into the repo safely.

Architecture in one view

Communication stores threads, participants, messages, and attachments. **AI Gateway** owns OpenAI calls, prompt and tool registries, parsing, tracing, and provider abstraction. **Studio** owns the structured design graph: requirements, decisions, entities, workflows, views, validations, and artifacts.

Layer	Primary responsibility
Communication	Threads, participants, messages, attachments, and reusable conversation history.
AI Gateway	OpenAI integration, prompt/tool registries, normalized outputs, request logs, and traceability.
Studio	Structured design state, validations, artifacts, and project/session orchestration.

Recommended usage order

- 1 Read this PDF first to understand the role of each file.

- 2 Review the Communication spec if you need shared thread/message infrastructure.
- 3 Review the AI Gateway spec if you need a clean OpenAI integration layer.
- 4 Review the Studio AI integration spec if you want Studio to use both layers without losing ownership of structured design data.
- 5 Use the Copilot install guide and install prompt when merging the files into your repo.
- 6 Convert the specs into `HN3T\_MASTER\_PLAN.md` sections only after the boundaries are accepted.

AI Gateway spec

This file defines the shared AI orchestration layer. It is where OpenAI integration should live so raw provider logic does not spread through Studio, Communication, or future modules.

- Use it to define provider accounts, model profiles, prompt templates, tool definitions, and request logs.
- Use it when deciding how to normalize AI outputs before they reach Studio or other modules.
- Do not use it as a replacement for RBAC, tenancy, or production business logic.

Communication spec

This file defines reusable interaction infrastructure for threads, participants, messages, attachments, and linked records. It should become the system-of-record for conversation history across Workforce.

- Use it when building module-linked discussions, AI conversation trails, approvals, or issue/task conversations.
- Keep it generic enough for reuse across multiple Workforce modules.
- Do not place Studio's structured design graph or raw OpenAI orchestration logic here.

Studio AI integration spec

This file defines the orchestration pattern that lets Studio use Communication for message persistence and AI Gateway for model interaction while Studio remains owner of requirements, decisions, workflows, and artifacts.

- Use it when designing the Studio message endpoint and backend orchestration flow.
- Use it to define normalized AI result contracts that Studio can persist safely.
- Do not allow provider-native payloads to become Studio's internal persistence contract.

Installation guidance

The install files are meant for merging these specs into an existing Workforce repo without replacing your current planning structure. Their job is to place the files into a planning tree, add links from your planning README, and preserve the boundary between Communication, AI Gateway, and Studio.

COPILOT_INSTALL_AI_COMM_STUDIO_SPEC_PACK.md

Use when you are installing the files manually or need a human-readable sequence.

COPILOT_INSTALL_AI_COMM_STUDIO_SPEC_PACK_PROMPT.md

Use when you want Copilot to add the files into the repo and keep the module boundaries explicit.

How this pack should be used inside Workforce

Treat this pack as source architecture documentation. After review, each spec should become one or more execution sections in `HN3T\_MASTER\_PLAN.md`, with Communication, AI Gateway, and Studio AI integration tracked as separate but connected workstreams.

Boundary rules to preserve

- Communication owns conversation transport and history, not Studio's structured design graph.
- AI Gateway owns OpenAI calls, prompts, tools, parsing, tracing, and provider abstraction.
- Studio owns structured design objects and remains the system of record for design state.
- OpenAI should help draft, extract, summarize, and validate, but Workforce decides what is persisted and what is allowed.
- All provider calls should remain server-side and scoped through Workforce authorization.

Suggested next deliverables

`AI\_GATEWAY\_\_ATOMIC\_EXECUTION\_PLAN.md` - turns the AI Gateway spec into a build sequence for provider config, request logs, prompt/tool registries, and tests.

`COMMUNICATION\_\_ATOMIC\_EXECUTION\_PLAN.md` - turns Communication into a reusable build plan for threads, participants, messages, attachments, and links.

`STUDIO\_AI\_INTEGRATION\_\_BUILD\_ORDER.md` - defines the rollout for linking Studio sessions to Communication threads and AI Gateway requests.

`HN3T\_MASTER\_PLAN.md` sections - lets the repo executor work through these pieces one atomic task at a time.

Quick install checklist

- Place the three spec files into your planning/specs directory.
- Link them from a planning README.
- Review and approve the module boundaries.
- Convert each spec into master-plan execution sections.
- Do not merge AI Gateway into Communication or Studio.
- Keep Communication reusable across future Workforce modules.

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